

External validation of the Khorana score for the prediction of venous thromboembolism in cancer patients: A systematic review and meta-analysis[☆]

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ABSTRACT

Background: Venous thromboembolism is the leading cause of death in cancer patients, second only to tumor progression. The Khorana score is recommended by clinical guidelines for identifying ambulatory cancer patients at high risk of venous thromboembolism during chemotherapy. However, its predictive performance is debated among cancer patients.

Objectives: To map the applicability of the Khorana score in cancer patients and to assess its predictive performance across various cancer types, providing guidance for clinicians and nurses to use it more appropriately.

Design: Systematic review and meta-analysis.

Methods: A comprehensive literature search of the electronic database was first conducted on August 30, 2023, and updated on May 20, 2024. Studies examining the Khorana score's predictive performance (including but not limited to the areas under the curve, C-index, and calibration plot) in cancer patients were included. The Prediction Model Risk of Bias Assessment Tool (PROBAST) was applied to evaluate the methodological quality of the included studies. Data synthesis was achieved via random-effects meta-analysis using the R studio software. The subgroup analysis was performed according to the study design, clinical setting, cancer type, anti-cancer treatment stage, and country.

Results: The review incorporated 67 studies, including 58 observational studies and nine randomized controlled trials. All included studies assessed the Khorana score's discrimination, with the C-index ranging from 0.40 to 0.84. The pooled C-index for randomized controlled trials was 0.61 (95 % CI 0.51–0.70), while observational studies showed a pooled C-index of 0.59 (95 % CI 0.57–0.60). Subgroup analyses revealed the pooled C-index for lung cancer, lymphoma, gastrointestinal cancer, and mixed cancer patients as 0.60 (95 % CI 0.53–0.67), 0.56 (95 % CI 0.51–0.61), 0.59 (95 % CI 0.39–0.76), and 0.60 (95 % CI 0.58–0.61), respectively. Inpatient and outpatient settings had the pooled C-index of 0.60 (95 % CI 0.58–0.63) and 0.58 (95 % CI 0.55–0.61), respectively. Calibration was assessed in only four studies. All included studies were identified to have a high risk of bias according to PROBAST.

Conclusion: The Khorana score has been widely validated in various types of cancer patients; however, it exhibited poor capability (pooled C-index < 0.7) in accurately discriminating VTE risk among most types of cancer patients either in inpatient or outpatient settings. The Khorana score should be used with caution, and high-quality studies are needed to further validate its predictive performance.

Registration: The protocol for this study is registered with PROSPERO (registration number: CRD42023470320).

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What is already known

- Precisely screening high-risk groups for venous thromboembolism in cancer patients and taking reasonable anticoagulation prophylaxis could halve the relative risk of VTE.

- Clinical guidelines recommend the Khorana score for identifying ambulatory cancer patients with a high risk of venous thromboembolism receiving chemotherapy.
- The predictive performance of the Khorana score in predicting the risk of venous thromboembolism in cancer patients is debated.

What this paper adds

- This study demonstrates that the Khorana score's application has been extended beyond its initial target population and settings.
- Although the Khorana score was developed based on mixed types of cancer patients, this study demonstrated that it failed to accurately identify high-risk patients for venous thromboembolism when applied to specific cancer populations, especially in patients with lung cancer, lymphoma, and gastrointestinal cancer.
- The Khorana score was also incapable of accurately discriminating the risk level of venous thromboembolism in both inpatient and outpatient settings for cancer patients, despite its original development for outpatient clinic use.

1. Introduction

As the foremost non-tumoral cause of mortality, venous thromboembolism (VTE), commonly encompassing deep vein thrombosis (DVT) and pulmonary embolism (PE), is a critical focus for clinicians and nurses (Brown et al., 2023; Guntupalli et al., 2023). VTE is an essential indicator in evaluating the quality of medical treatment and clinical nursing (Goldsmith et al., 2008). The reported incidence of cancer-related VTE is as high as 23.11 % (Martín et al., 2016), a rate sevenfold higher than that of non-cancer cohorts (Schrag et al., 2023), accompanied by a mortality rate that is disproportionately elevated 16 times higher than their non-cancer counterparts (Puurunen et al., 2016). Additionally, VTE could interrupt anti-cancer treatment progress and increase mortality risk and related medical costs (Khorana et al., 2013).

The Khorana score is one of the most widely used risk assessment tools for predicting VTE in ambulatory cancer patients receiving chemotherapy, as supported by numerous studies and clinical guidelines (Khorana et al., 2008; Key et al., 2020; Farge et al., 2022; Streiff et al., 2015, 2021; Ma et al., 2019; Mosarla et al., 2019). It incorporates five easily accessible variables: cancer site, platelet count, hemoglobin and/or use of erythropoiesis-stimulating agents, leukocyte count, and body mass index. The score was originally developed based on data from various types of cancer patients and aimed to predict the VTE risk in cancer patients receiving chemotherapy in outpatient clinics. Its predictive performance has currently been evaluated in cohorts with certain specific cancer types (Sanfilippo et al., 2022; Oto et al., 2020; Iqbal et al., 2021) and across a wider range of clinical settings (Michalski et al., 2023; Jin et al., 2022).

However, the predictive performance of the Khorana score has varied significantly among different types of cancer patients. For example, a study by Haas et al. reported an area under the curve (AUC) of 0.40 for the Khorana score in breast cancer patients (Haas et al., 2012). In contrast, a study of lung cancer patients reported an AUC of 0.74 for the Khorana score (Vathiotis et al., 2018). Moreover, the predictive performance of the Khorana score has been controversial even within the same type of cancer. For instance, in lung cancer patients, several studies showed the C-index of approximately 0.50 of the Khorana score (Sanfilippo et al., 2022; Rupa-Matysek et al., 2017), while others reported AUC values greater than 0.70 (Vathiotis et al., 2018; Li et al., 2019; S. Li et al., 2021). Existing systematic reviews, while informative, have either focused exclusively on VTE incidence in different risk groups categorized by the Khorana score or were limited to randomized controlled trials (Mulder et al., 2019; Van Es et al., 2020), underscoring the need for a more comprehensive evaluation of the Khorana score.

Nurses play an essential role in VTE prevention, as they are responsible for using tools to evaluate patients' VTE risk. It is the nurse's

responsibility to ensure that the chosen risk assessment tool is appropriate for the specific clinical setting. Therefore, this study aimed to address these gaps by performing a systematic review with a broader scope, offering a nuanced assessment of the Khorana score's applicability and predictive performance across diverse cancer types and clinical settings, and identifying its optimal use scenarios.

2. Methods

This systematic review was reported according to the Transparent Reporting of Multivariable Prediction Models for Individual Prognosis or Diagnosis: Checklist for Systematic Reviews and Meta-analyses (TRIPOD-SRMA) and was registered with PROSPERO (registration number: CRD42023470320). The TRIPOD-SRMA checklists for abstract and main text are shown in Supplementary Tables 1 and 2.

2.1. Search strategy

We systematically searched the databases of PUBMED, Web of Science, EMBASE, Cumulative Index to Nursing and Allied Health Literature (CINAHL), the Cochrane Library, SinoMed, China Science and Technology Journal Database (VIP), China National Knowledge Infrastructure (CNKI), and Wanfang Database from inception to 30 August 2023. Additionally, an updated literature search was conducted on May 20, 2024 to identify newly published eligible studies. The search terms included Khorana risk assessment, Khorana score, Khorana model, Khorana RAM, Khorana risk tool, Khorana scales, and KRS. Detailed search strategies are shown in Supplementary Table 3. In addition, we manually searched the reference list of eligible studies and related reviews, as well as the studies that cited the original study developing the Khorana score to include potentially eligible studies.

2.2. Eligibility criteria

Eligibility criteria were established based on the PICOTS framework, as detailed in Table 1. The specific inclusion criteria of this study were (1) studies that externally validated the predictive performance of the Khorana score, (2) studies involving patients diagnosed with cancer, and (3) studies published in English or Chinese. Studies would be excluded if they included non-cancer patients or if the required information on predictive performance cannot be extracted. In instances of multiple publications reporting on an identical dataset, the earliest published article was selected for inclusion. Meeting abstracts that reported on the predictive performance of the Khorana score were also considered for inclusion.

2.3. Studies selection

The selection process of studies was performed simultaneously and independently by two authors (XP and LY). First, the Endnote X9 software was used to find and remove duplicates. Then, the title and abstract were screened to eliminate the studies that did not meet the inclusion criteria. Finally, the full text of the remaining articles was read carefully to identify

Table 1
The detailed description of the PICOTS for this systematic review.

Item	Definition
Population	Patients diagnosed with cancer
Intervention	The Khorana score
Comparator	Not applicable
Outcomes	Objectively diagnosed VTE, including but not limited to the type of deep vein thrombosis, pulmonary embolism, and central venous catheter-related thrombosis
Timing	Predictors were measured at any time point preceding the occurrence of VTE; the outcome (VTE) was measured without any specific limitation in the prediction horizon
Setting	Not limited to any specific clinical setting

VTE: venous thromboembolism.

the eligible studies. In case of disagreements regarding study selection, a discussion involving three authors (WG) was held to reach a consensus.

2.4. Data extraction

To facilitate the data extraction process, we developed a standardized form according to the checklist for critical appraisal and data extraction for systematic reviews of prediction modeling studies (CHARMS). Two authors (HX and CHX) independently extracted related data, including first author, country, study design, setting, sample size, cancer type of study population, type and incidence of VTE, timing of predictor measurement, over what period the VTE was predicted, anti-cancer treatment stage, and model predictive performance (i.e., discrimination and calibration) of the Khorana score. We chose discrimination and calibration because these are the two most important and common indicators of model predictive performance (Debray et al., 2017). Potential measures of discrimination include the C-index (equal to the area under the curve in logistic regression) and D-statistic, and potential measures of calibration include the calibration plot, calibration slope, Hosmer–Lemeshow test, and total ratio of observed (O) event to expected (E) event (O:E ratio) (Debray et al., 2017). Any discrepancies between authors were resolved through discussion or by a third author (PLH).

2.5. Quality appraisal

Two authors (HX and MS) independently evaluated the risk of bias and the applicability of the included studies by the Prediction Model Risk of Bias Assessment Tool (PROBAST) (Moons et al., 2019) was applied to by two authors. The PROBAST encompassed four domains of Participants, Predictors, Outcome, and Analysis for risk of bias assessment, and three domains of Participants, Predictors, and Outcomes for applicability assessment. The domain of participants mainly assesses the potential sources of bias regarding the data source and inclusion and exclusion criteria with two questions. The predictors domain assesses the potential sources of bias related to the definition and measurement of predictors with three questions. The outcome domain assesses the potential sources of bias related to the definition and determination of the outcome with five questions. The analysis domain examines whether key statistical considerations (e.g., the sample size calculation methods, methods for handling independent variables and missing data, predictor selection methods, and model predictive performance assessments) are correctly addressed with nine questions. Each domain's questions are answered with yes, probably yes, probably no, no, or no information, depending on the exact characteristics of each study. A domain is deemed at high risk of bias if it had ≥ 1 question answered with 'no' or 'probably no.' Conversely, a domain was considered at low risk of bias only if all questions were answered with 'yes' or 'probably yes.' A domain received an 'unclear risk of bias' designation if one or more questions were answered with 'no information,' with the remaining questions answered with 'yes' or 'probably yes.' The criteria for overall assessment of risk of bias and applicability for individual studies are as follows: (1) If all domains are rated low risk of bias/low concerns for applicability, the overall judgment of the individual study will be low risk of bias/low concerns for applicability. (2) If ≥ 1 domain is rated to be at a high risk of bias/high concerns for applicability, the overall judgment will be a high risk of bias/high concerns for applicability. (3) If ≥ 1 domain is rated unclear risk of bias/unclear concern for applicability and all other domains are rated low risk of bias/low concern for applicability, the overall judgment will be unclear risk of bias/unclear concerns of applicability. Discrepancies between authors HX and MS were resolved through discussion or by author PLH.

2.6. Statistical analysis

Data analyses were conducted using R Studio for MacOS. The frequency and proportion were used to describe the general characteristics of the included studies. The discrimination of the Khorana score was

appraised by weighting the C-index/AUC of each study by its sample size and synthesizing the data using the 'metamisc' and 'metafor' packages with a random effects model. The C-index, ranging from 0.5 (no discrimination) to 1.0 (perfect discrimination), is generally accepted as a good discriminative ability with C-index ≥ 0.70 (Yourman et al., 2012). Multiple methodologies, including calibration plots, calibration slopes, and the Hosmer–Lemeshow test, are commonly used to assess calibration performance, yet synthesizing these measurements proved challenging (Debray et al., 2017). Debray et al. (2017) recommended synthesizing the observed-to-expected event (O:E) ratio to roughly estimate the model's calibration. However, the O:E ratio was not reported in any included study, and it could not be estimated by the formula provided by Debray et al. because of the lack of required information. Consequently, only a qualitative analysis of calibration performance was feasible. Subgroup analyses were conducted on study design, cancer type, clinical setting, anti-cancer treatment stage, and country if there were two or more original studies. To evaluate the robustness of the synthesized findings, sensitivity analyses were performed by sequential exclusion of individual studies. Heterogeneity was assessed by calculating the I^2 statistic, with values of 25 %, 50 %, and 75 % indicating low, moderate, and high heterogeneity, respectively (Huedo-Medina et al., 2006). The Egger test and funnel plot were used to evaluate publication bias, and $p > 0.05$ indicated a low possibility of publication bias (Egger et al., 1997).

3. Results

3.1. Search results and overview of included studies

A total of 1649 articles were retrieved from electronic databases, and 24 potentially eligible studies were retrieved manually. After reading the title, abstract, and full text, 61 studies were included. An updated search was carried out on May 20, 2024 and additional six studies were included. As a result, 67 studies were included for final analysis, with 75 datasets (Fig. 1).

Geographically, the validation cohorts predominantly came from China ($n = 23$), followed by the United States ($n = 9$), Spain ($n = 7$), and the Netherlands ($n = 5$). More than half of the included studies were retrospective cohort or case–control studies ($n = 42$), 16 studies were prospective cohort studies, and nine studies were randomized controlled trials. The sample sizes of the included studies ranged from 38 to 40,218 participants. Notably, a substantial proportion of these studies ($n = 46$, 68.7 %) comprised sample sizes of fewer than 1000 patients, as detailed in Table 2.

3.2. The applicability status of the Khorana score (Table 2)

Approximately half of the studies ($n = 34$, 50.8 %) lacked explicit reporting on the clinical setting where the Khorana score was used. Of the remaining studies, 12 studies validated the score within outpatient clinics, while 21 conducted validation in inpatient settings. Among the included studies, 27 involved cancer patients undergoing chemotherapy, 6 included surgical cancer patients, and 34 either did not specify the anti-cancer treatment received by participants or included diverse anti-cancer treatment modalities.

Despite its origin in a mixed cancer patient cohort, the Khorana score has been assessed in specific cancer types, notably lung cancer ($n = 12$), with validations also conducted across a spectrum of other malignancies, such as lymphoma ($n = 5$), colorectal cancer ($n = 3$), germ cell cancer ($n = 3$), gastrointestinal cancer ($n = 2$), multiple myeloma ($n = 2$), glioma ($n = 2$), and breast cancer ($n = 2$). Furthermore, 33 studies undertook evaluations within mixed cancer patient cohorts.

For predictor measurement timing, 30 studies failed to report specifics, particularly regarding laboratory parameters such as platelet, hemoglobin, and leukocyte counts. Twenty-nine studies did not delineate the timeframe over which VTE was predicted. In studies that

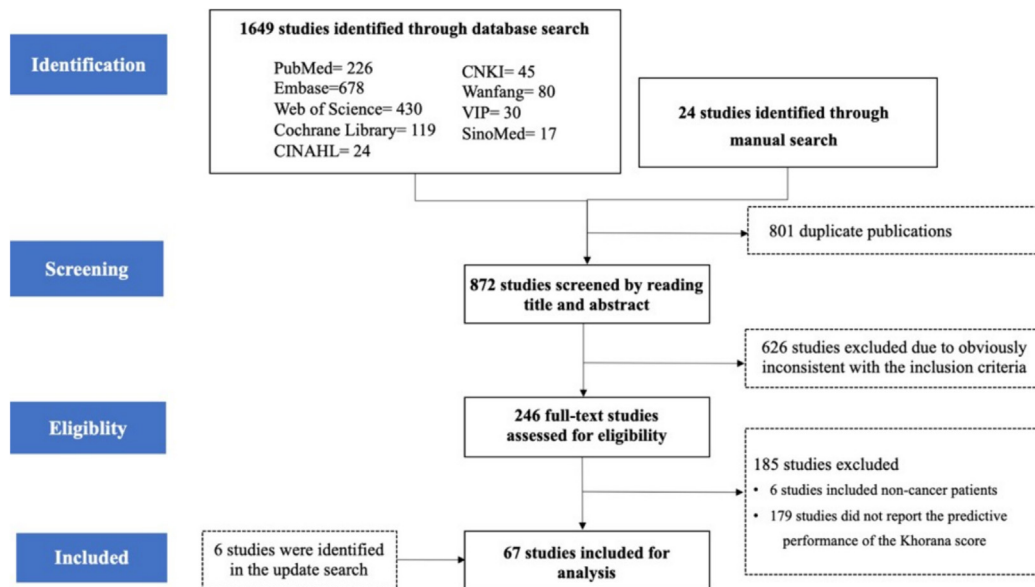


Fig. 1. Flowchart of the study selection procedure.

specified a prediction period, 12 used the Khorana score for forecasting VTE risk within six months post-chemotherapy initiation.

Predicted VTE types across studies varied, encompassing DVT ($n = 45$), PE ($n = 37$), central venous catheter-related thrombosis ($n = 6$), superficial vein thrombosis ($n = 6$), and other VTE in rare sites such as renal, retinal, cerebral, or portal veins. Details of the included studies are presented in Table 2.

3.3. Discrimination of the Khorana score

The C-index of the Khorana score ranged markedly from 0.40 to 0.84. Notably, 59 studies reported a C-index below the threshold of 0.70. The pooled C-index for observational studies was 0.59 (95 % CI 0.57–0.60), while for randomized controlled trials, it was slightly higher at 0.61 (95 % CI 0.51–0.70), as illustrated in Fig. 2.

The subgroup analysis delineated by cancer type showed that, within observational studies, patients with germ cell cancer got the highest pooled C-index (0.61, 95 % CI 0.53–0.69), followed by mixed cancer (0.60, 95 % CI 0.58–0.61), lung cancer (0.60, 95 % CI 0.53–0.67), gastrointestinal cancer (0.59, 95 % CI 0.39–0.76), lymphoma (0.56, 95 % CI 0.51–0.61), colorectal cancer (0.56, 95 % CI 0.47–0.65), and multiple myeloma (0.53, 95 % CI 0.53–0.53). For randomized controlled trials, subgroup analyses were conducted solely on mixed cancer and lung cancer, yielding the pooled C-index of 0.68 (95 % CI, 0.44–0.86) and 0.51 (95 % CI, 0.46–0.56), respectively. Details are shown in Fig. 3.

The subgroup analysis performed in clinical setting showed that the synthesized C-index was 0.60 (95 % CI 0.58–0.63) for inpatient settings and 0.58 (95 % CI 0.55–0.61) for outpatient settings. Details are shown in Fig. 4.

The subgroup analysis performed on anti-cancer treatment stage showed that the pooled C-indices for chemotherapy, surgery, and other unspecified treatment were 0.57 (95 % CI 0.55–0.59), 0.60 (95 % CI 0.53–0.66), and 0.60 (95 % CI 0.57–0.62), respectively. Details are available in Supplementary Fig. 1.

The subgroup analysis performed on country showed that the pooled C-indices of populations from Asian, North American, European, and Oceanian countries were 0.61 (95 % CI 0.58–0.64), 0.60 (95 % CI 0.57–0.63), 0.56 (95 % CI 0.55–0.58), and 0.57 (95 % CI 0.18–0.89), respectively. Results about other countries are shown in Supplementary Fig. 2. Sensitivity analyses were performed by sequential exclusion of

individual studies showed that the aforementioned synthesized results did not change significantly, indicating the stability of the results.

3.4. Calibration validation of the Khorana score

Only four studies reported the results about the calibration which were evaluated using calibration plots (Hu et al., 2020; Jin et al., 2022; Gomez-Rosas et al., 2023; Yang et al., 2024). Among these four studies, only one study demonstrated an excellent fit between predicted probabilities and observed probabilities of VTE (Hu et al., 2020).

3.5. Risk of bias and applicability concerns based on PROBAST

Upon application of the PROBAST tool, as shown in Table 3, all included studies were deemed to exhibit a high risk of bias. A majority of the studies ($n = 64$, 95.5 %) incurred a high risk of bias in the Analysis domain, predominantly attributed to the absence of calibration evaluation. Concerning applicability, only seven studies were considered to have a high level of concern. The majority of studies were classified as having uncertain ($n = 43$) or low ($n = 17$) concerns regarding applicability. The summarized results of the risk of bias and applicability assessment for all included studies were provided in Supplementary Fig. 3.

3.6. Publication bias

The funnel plot demonstrated a symmetric distribution (Supplementary Fig. 4), suggesting no evident publication bias. The Egger test yielded a non-significant bias coefficient ($p = 0.076$), further supporting the results yielded from the funnel plot (Supplementary Table 4).

4. Discussions

This systematic review and meta-analysis summarized and analyzed the application status and the predictive performance of the Khorana score. The key findings of this study are as follows: (1) The Khorana score has been evaluated across various cancers, including but not limited to lung cancer, lymphoma, and gastrointestinal cancer. Notwithstanding, there remains a paucity of evaluation within certain oncological contexts; (2) the Khorana score demonstrated limited discriminative ability for VTE in both mixed cancer cohorts and cohorts of patients with specific cancer

Table 2
Overview of the included studies evaluating the predictive performance of the Khorana score (n = 67).

ID	Author(s) (year)	Design	Setting	Sample size	Anti-cancer treatment phase	Predictor measurement timepoint	Predicted timespan	Cancer type	VTE type	VTE incidence	Discrimination
1	Agnelli et al. (2012)	RCT	—	3212	Chemotherapy	After chemotherapy initiation	3.5 months after the start of chemotherapy	Mixed	VTE	2.33 %	AUC = 0.56
2	Alexander et al. (2019)	PS	—	188	—	Pre-chemotherapy	—	Lung	DVT, PE	12.77 %	AUC = 0.51
3	Bamias et al. (2020)	RS	Inpatient	335	Chemotherapy	Pre-chemotherapy	3.1 months after the start of chemotherapy	Lung Urothelium	DVT, PE	8.66 %	AUC = 0.58
4	Bastos-Oreiro et al. (2021)	RS	—	208	—	After diagnosis	6 months after diagnosis	Lymphoma	DVT, PE, CRT	14.90 %	AUC = 0.50 Sen = 0.06 Spe = 0.94 PPV = 0.17 NPV = 0.84
5	Brown et al. (2024)	RS	—	123	Chemotherapy	—	6 months after chemotherapy	Germ cell	DVT, PE, CRT	26.02 %	AUC = 0.68
6	Cella et al. (2017)	PS	—	843	—	—	—	Mixed	VTE	8.65 %	AUC = 0.58
7	Chen et al. (2022)	PS	Inpatient	491	Surgery	Before discharge	Within 1 month after discharge	Breast	DVT	2.24 %	AUC = 0.58
8	Dharmavaram et al. (2020)	RS	—	790	Other treatment	—	1 year after diagnosis	Lymphoma	DVT, PE	13.42 %	Training cohort: AUC = 0.60 Validation cohort: AUC = 0.61
9	Di et al. (2023)	PS	Inpatient	158	—	—	6 months after diagnosis	Lung	VTE	17.09 %	AUC = 0.53
10	Di Nisio et al. (2019)	RS	—	776	Chemotherapy	Pre-chemotherapy	1 year after the start of chemotherapy	Mixed	DVT, PE, SVT	8.63 %	AUC = 0.61
11	Douce et al. (2017)	RS	—	1583	Chemotherapy	Pre-chemotherapy	6 months after the start of chemotherapy	Mixed	VTE	11.81 %	C-index = 0.61
12	Edwin et al. (2016)	RS	—	1520	Other treatment	Before anti-cancer therapy	6 months after diagnosis	Multiple myeloma	VTE	6.78 %	C-index = 0.53
13	Feroni et al. (2017)	RS	Outpatient	608	Chemotherapy	Pre-chemotherapy	—	Mixed	VTE	7.10 %	AUC = 0.59 Sen = 0.09 Spe = 0.94
14	Gomez-Rosas et al. (2023)	RS	—	568	Chemotherapy	Pre-chemotherapy	—	Lung	DVT, PE	10.92 %	AUC = 0.54 Sen = 0.21 Spe = 0.86
15	Guman et al. (2023)	PS	—	36,150	—	—	12 months after diagnosis	Mixed	VTE	2.82 %	C-index = 0.60
16	Guman et al. (2021b)	RS	—	2729	—	Pre-chemotherapy	6 months after the start of chemotherapy	Mixed	DVT, PE	5.86 %	C-index = 0.57 Sen = 0.17
17	Guman et al. (2021a)	RS	—	299	Chemotherapy	—	6 months after cancer diagnosis	Mixed	DVT, PE, SVT	7.36 %	C-index = 0.55
18a	Haas et al. (2012)	RCT	—	532	Chemotherapy	—	6 months after the start of chemotherapy	Lung	DVT, PE, SVT	13.53 %	AUC = 0.54
18b	Haas et al. (2012)	RCT	—	351	Chemotherapy	—	6 months after the start of chemotherapy	Breast	DVT, PE, SVT	8.26 %	AUC = 0.40
19	Hu et al. (2020)	RS	Inpatient	442	—	—	—	Mixed	DVT, PE	50.00 %	AUC = 0.58 Sen = 0.35 Spe = 0.79
20	Huang (2022)	PS	Inpatient	2067	—	First admission after cancer diagnosis	Within 1 year after cancer diagnosis	Mixed	DVT, PE, SVT, rare VTE	21.09 %	AUC = 0.53
21	Iqbal et al. (2021)	RS	Inpatient	589	Surgery	—	Within 6 months after surgery	Bladder	DVT, PE	5.60 %	AUC = 0.51
22	Jin et al. (2022)	RS	Inpatient	1035	—	—	—	Mixed	DVT, PE, rare site VTE	22.30 %	AUC = 0.60 Sen = 0.39
23	Khan et al. (2022)	RS	Outpatient	270	—	—	—	Mixed	VTE	—	Spe = 0.76
24	Kuang et al. (2019)	RS	Inpatient	504	—	Pre-chemotherapy	—	Mixed	DVT	12.50 %	AUC = 0.66
25	Lecumberri et al. (2013)	RCT	—	38	Chemotherapy	After chemotherapy initiation	—	Lung	VTE	10.53 %	AUC = 0.65 AUC = 0.77
26	Li et al. (2023)	RS	—	21,142	Other treatment	—	6 months after the start of treatment	Mixed	DVT, PE	5.70 %	C-index = 0.64
27	S. Li et al. (2021)	PS	—	124	—	—	—	Lung	VTE	19.36 %	AUC = 0.71

(continued on next page)

Table 2 (continued)

ID	Author(s) (year)	Design	Setting	Sample size	Anti-cancer treatment phase	Predictor measurement timepoint	Predicted timespan	Cancer type	VTE type	VTE incidence	Discrimination
28	Li (2020)	RS	Inpatient	3730	Other treatment	Within 48 h of admission	—	Mixed	DVT, PE	4.21 %	AUC = 0.74
29	Li et al. (2019)	RS	Inpatient	63	—	—	—	Lung	DVT, PE	33.33 %	AUC = 0.74 Sen = 0.62 Spe = 0.81
30	Liu et al. (2020)	RS	—	3877	Chemotherapy	Pre-chemotherapy	3 or 12 months after the start of treatment	Mixed	CRT	12.97 %	Development cohort: AUC = 0.54Validation cohort:
31	Macbeth et al. (2016)	RCT	—	2202	Other treatment	After chemotherapy initiation	—	Lung	VTE	7.90 %	AUC = 0.55 AUC = 0.51
32	Martín et al. (2016)	PS	Outpatient	251	Chemotherapy	—	6 months after the start of chemotherapy	Mixed	VTE	23.11 %	AUC = 0.55 Sen = 0.64 Spe = 0.47
33	Michalski et al. (2023)	RS	Inpatient	495	Chemotherapy	After chemotherapy initiation	6 months after the end of first-line chemotherapy	Germ cell	DVT, PE	13.94 %	AUC = 0.59
34	Munoz et al. (2023)	PS	Outpatient	627	—	—	6 months after diagnosis	Mixed	VTE	15.15 %	Development cohort: AUC = 0.58 Sen = 0.26 Spe = 0.82Validation cohort: AUC = 0.58 Sen = 0.20 Spe = 0.89
35	Muñoz Martín et al. (2018)	RS	Outpatient	391	Chemotherapy	Pre-chemotherapy	6 months after diagnosis	Mixed	VTE	18.16 %	AUC = 0.58 Sen = 0.22 Spe = 0.82 NPV = 0.22 PPV = 0.82
36	Niimi et al. (2024)	RS	—	208	—	—	—	Mixed	DVT	28.37 %	NPV = 0.82 AUC = 0.61
37	Oto et al. (2020)	PS	—	50	Surgery	—	—	Glioma	PE	34.00 %	AUC = 0.52
38	Overvad et al. (2022)	PS	Outpatient	40,218	Chemotherapy	Pre-chemotherapy	6 months after the start of chemotherapy	Mixed	DVT, PE rare VTE	2.49 %	AUC = 0.60
39	Pabinger et al. (2018)	PS	—	2255	—	—	6 months after diagnosis	Mixed	DVT	5.68 %	Development cohort: AUC = 0.61Validation cohort:
40	Pelzer et al. (2015)	RCT	—	312	Chemotherapy	After chemotherapy initiation	3 months after the start of chemotherapy	Pancreatic	DVT	5.45 %	AUC = 0.56 AUC = 0.61
41	Perry et al. (2010)	RCT	—	186	Surgery	—	6 months after surgery	Glioma	DVT, PE	11.83 %	AUC = 0.63
42	Qin et al. (2023b)	RS	Inpatient	1191	Surgery	Prior surgery	30 days after surgery	Colorectal	CRT, PE	10.83 %	AUC = 0.65 Sen = 0.11 Spe = 0.98 PPV = 0.40 NPV = 0.90
43	Qin et al. (2023a)	RS	Inpatient	1392	—	—	—	Colorectal	DVT, PE	35.63 %	AUC = 0.55
44	Ramos-Esquivel et al. (2023)	RS	Outpatient	708	Chemotherapy	Pre-chemotherapy	6 months after the start of chemotherapy	Mixed	DVT	4.24 %	C-index = 0.62 Sen = 0.69 Spe = 0.60
45	Rupa-Matysek et al. (2018)	RS	—	167	Other treatment	Pre-chemotherapy	1 month after the start of treatment	Lymphoma	DVT, PE	7.19 %	AUC = 0.52 Sen = 1.00 Spe = 0.00 PPV = 1.00

46	Rupa-Matysek et al. (2017)	RS	Outpatient	428	Chemotherapy	Pre-chemotherapy	2.5 months after the start of chemotherapy	Lymphoma	DVT, PE	14.95 %	NPV = 0.00 C-index = 0.51 Sen = 0.17 Spe = 0.85 PPV = 0.15 NPV = 0.82 C-index = 0.53
47	Sanfilippo et al. (2022)	RS	—	2870	Chemotherapy	Pre-chemotherapy	Within the first 6 months of anti-treatment	Multiple myeloma	DVT, PE	4.46 %	
48	Shibata et al. (2023)	RS	Inpatient	250	Chemotherapy	—	1 year after the start of chemotherapy	Gastrointestinal	DVT, PE	24.40 %	AUC = 0.49
49	Shoji et al. (2024)	RS	Outpatient	9965	—	Pre-chemotherapy	—	Mixed	VTE	1.97 %	AUC = 0.65
50	Srikanthan et al. (2015)	RS	Inpatient	324	Chemotherapy	Pre-chemotherapy	3 months after the start of chemotherapy	Germ cell	DVT, PE	9.57 %	Training cohort: AUC = 0.67
51	Sun (2019)	RCT	—	105	—	—	—	Mixed	VTE	21.90 %	Validation cohort: AUC = 0.57
52	Thipbumrung et al. (2023)	PS	Outpatient	457	Chemotherapy	After chemotherapy	6 months after chemotherapy	Mixed	DVT, PE	6.56 %	AUC = 0.84 C-index = 0.57 Sen = 0.23 Spe = 0.90 AUC = 0.53
53	Trujillo-Santos et al. (2021)	PS	Outpatient	1654	—	—	—	Mixed	VTE	7.80 %	
54	Tsubata et al. (2023)	PS	—	1008	—	—	2 years after diagnosis	Lung	DVT, PE	9.92 %	AUC = 0.52 Sen = 0.06 C-index = 0.52
55	Van Es et al. (2017)	PS	Outpatient	876	Chemotherapy	Pre-chemotherapy	6 months after the start of chemotherapy	Mixed	DVT, PE	6.05 %	
56	Vathiotis et al. (2018)	RS	—	130	Chemotherapy	—	—	Lung	DVT, PE	10.00 %	AUC = 0.74
57	M.M. Wang et al. (2019)	RS	Inpatient	361	—	—	—	Mixed	DVT	63.71 %	AUC = 0.60 Sen = 0.81 Spe = 0.37 AUC = 0.64 Training set: AUC = 0.52 Validation set: AUC = 0.54
58	X. Wang et al. (2019)	RS	Inpatient	212	—	—	—	Gynecological	DVT	50.00 %	
59	Yang et al. (2024)	RS	—	1996	—	—	—	Colon	DVT, PE	9.12 %	
60	Yang et al. (2021)	RS	—	555	Chemotherapy	Pre-chemotherapy	—	Lymphoma	DVT	20.36 %	AUC = 0.56
61	Yuen et al. (2022)	RS	—	298	Chemotherapy	Pre-chemotherapy	—	Mixed	DVT PE, CRT	7.14 %	AUC = 0.59
62	Zhang et al. (2022b)	RS	Inpatient	369	Other treatment	After cancer diagnosis	6 months after diagnosis	Lung	DVT, SVT, PE	23.33 %	AUC = 0.60
63	Zhang et al. (2022a)	RS	Inpatient	1576	Other treatment	Before anti-cancer therapy	—	Gastrointestinal	DVT, PE, SVT	34.01 %	Training cohort: AUC = 0.59 Validation cohort: AUC = 0.66
64	Zhang and Lin (2019)	RS	—	150	Surgery	—	—	Lung	CRT	12.00 %	AUC = 0.64 Sen = 0.72 Spe = 0.62 AUC = 0.72
65	Zhang (2018)	RS	—	208	Other treatment	At diagnosis or before therapy	—	Mixed	DVT, PE	50.00 %	
66	Zhang (2016)	RCT	Inpatient	177	Chemotherapy	One cycle after chemotherapy	6 months after one cycle of chemotherapy	Mixed	VTE	19.21 %	AUC = 0.72
67	Zhong (2021)	RCT	Inpatient	90	—	Before anti-cancer therapy	—	Mixed	DVT, PE	81.11 %	AUC = 0.58

“—”, not reported or unable to judge based on existing information; RS, retrospective study; PS, prospective study; RCT, randomized controlled trial; DVT, deep vein thrombosis; PE, pulmonary thromboembolism; CRT, central venous catheter-related thrombosis; SVT, superficial vein thrombosis; AUC, areas under the receiving operator characteristic curve; Sen, sensitivity; Spe, specificity; NPV, negative predictive value; PPV, positive predictive value.

18a and 18b are data for two cancer types in one study. The DVT in this study refers to thrombosis of the deep vein system of upper and lower extremities, internal jugular vein, subclavian vein, superior vena cava, and inferior vena cava; CRT refers to central venous catheter-related thrombosis, peripherally inserted central catheter-thrombosis, and port-related thrombosis; rare site VTE refers to the thrombosis occurring in renal vein, cerebral vein, portal vein, and retinal vein.

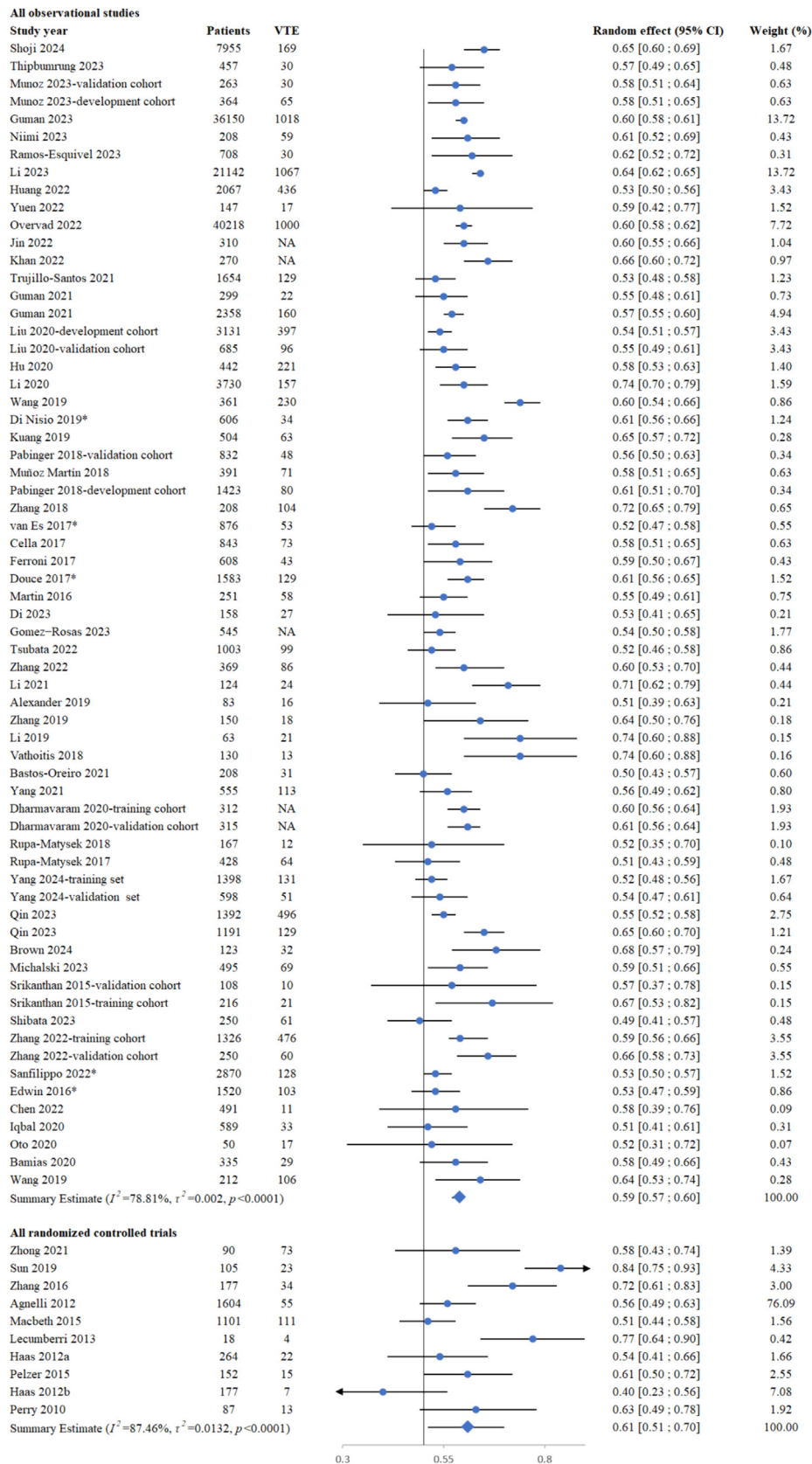


Fig. 2. Forest plot of pooled C-index estimates of all included studies. Note: *Select the C-index value obtained at the 6-month follow-up for analysis.

types; (3) the application setting of the Khorana score has expanded from outpatient chemotherapy to surgical hospitalization and other unspecified scenarios. However, subgroup analyses revealed poor discriminative

abilities of the Khorana score in both its original and extended settings; and (4) calibration evaluation of the score was markedly absent, indicating a need for further validation studies.

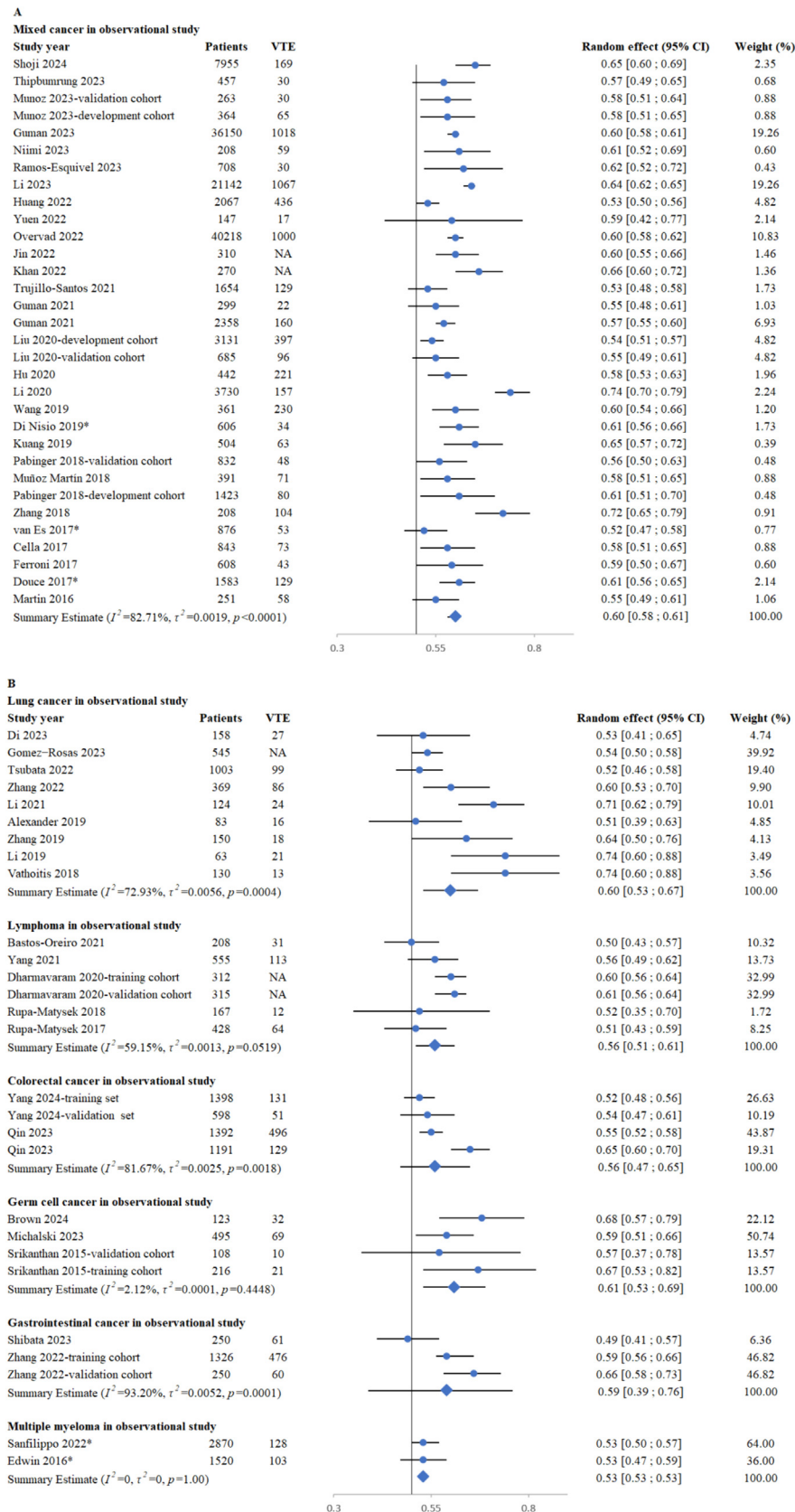


Fig. 3. (continued)

C

Mixed cancer in Randomized controlled trial

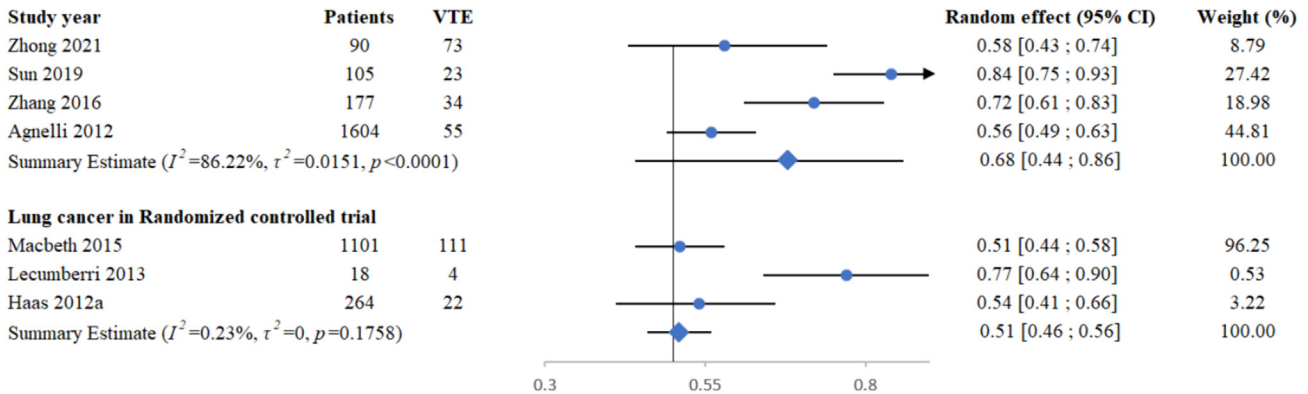


Fig. 3. Forest plot of subgroup analysis categorized by cancer types. A. Forest plot of mixed cancer in observational studies; B, forest plot of specific cancer in observational studies; C, forest plot of randomized controlled trials. Note: *Select the C-index value obtained at the 6-month follow-up for analysis.

In this study, we found that the Khorana score has been externally validated in twelve specific cancer types, with subgroup analyses performed on six cancer types (i.e., lung cancer, lymphoma, colorectal cancer, germ cell cancer, gastrointestinal cancer, and multiple myeloma). However, the pooled C-indices for these cancer types ranged from 0.51 to 0.61, which is lower than 0.7, indicating poor discriminative

ability among these cancer types. For cancer types with insufficient numbers of original studies to perform subgroup analysis (i.e., breast cancer, bladder cancer, urothelium cancer, glioma, gynecological cancer, and pancreatic cancer), the C-indices ranged from 0.40 to 0.64, also below 0.7. These results suggest that clinicians and nurses should be cautious when using the Khorana score for patients with the mentioned

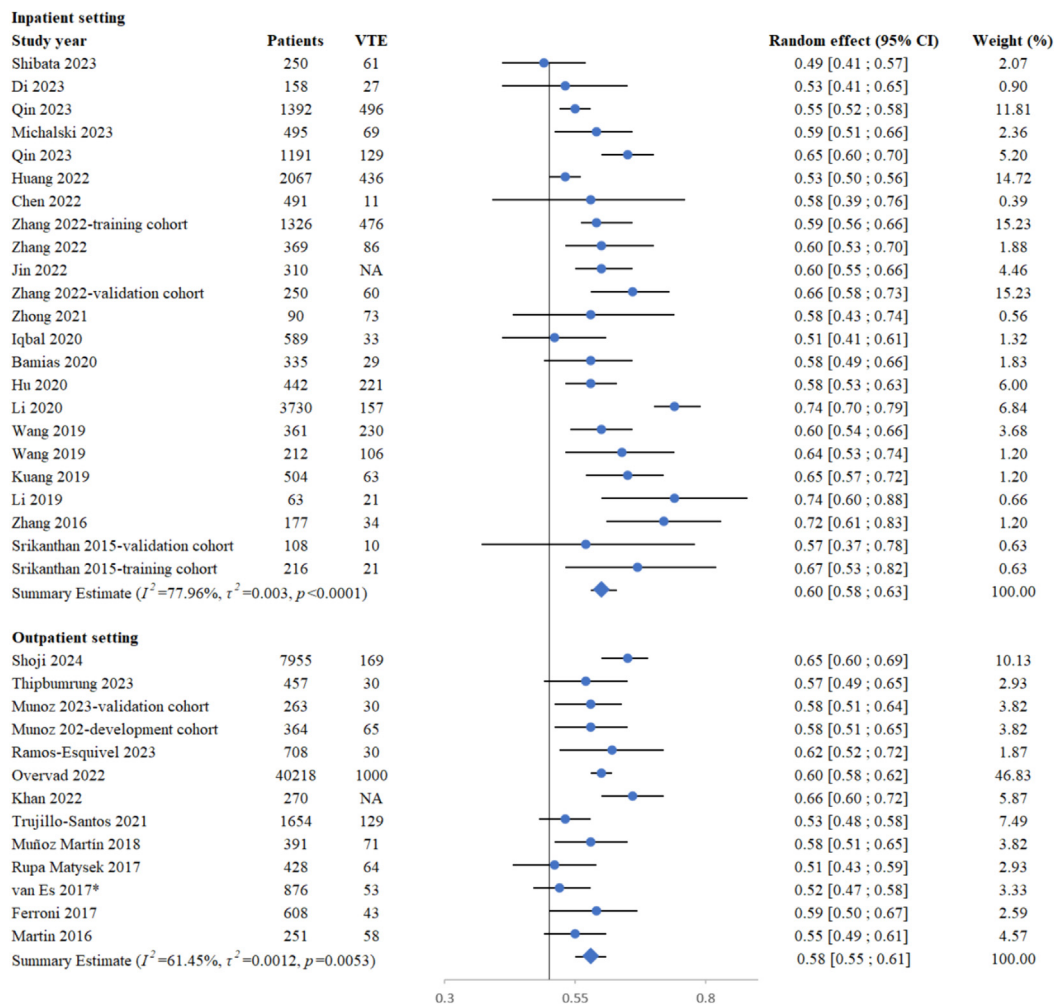


Fig. 4. Forest plot of pooled C-index estimates of clinical settings. Note: *Select the C-index value obtained at the 6-month follow-up for analysis.

Table 3

The detailed results of the risk of bias and applicability assessment by PROBAST for all included studies.

Study	Applicability				Risk of bias				
	Participants	Predictors	Outcome	Overall	Participants	Predictors	Outcome	Analysis	Overall
Agnelli et al. (2012)	—	+	—	+	—	—	?	+	+
Alexander et al. (2019)	—	—	?	?	?	—	—	+	+
Bamias et al. (2020)	—	—	—	—	+	—	—	+	+
Bastos-Oreiro et al. (2021)	—	—	—	—	+	—	—	+	+
Brown et al. (2024)	—	—	—	—	+	—	—	+	+
Cella et al. (2017)	—	?	?	?	—	—	?	+	+
Chen et al. (2022)	—	+	—	+	—	—	—	+	+
Dharmavaram et al. (2020)	—	?	—	?	+	—	?	+	+
Di et al. (2023)	—	—	—	—	—	—	—	+	+
Di Nisio et al. (2019)	—	—	—	—	+	—	—	+	+
Douce et al. (2017)	—	—	—	—	+	—	—	+	+
Edwin et al. (2016)	—	—	—	—	+	—	—	+	+
Ferroni et al. (2017)	—	—	?	?	+	—	—	+	+
Gomez-Rosas et al. (2023)	—	?	?	?	+	—	?	+	+
Guman et al. (2023)	—	?	—	?	—	—	?	+	+
Guman et al. (2021b)	—	?	?	?	—	—	—	+	+
Guman et al. (2021a)	—	—	—	—	?	—	?	+	+
Haas et al. (2012)	—	?	—	?	?	—	?	+	+
Hu et al. (2020)	—	?	?	?	+	—	?	—	+
Huang (2022)	—	—	—	—	—	—	—	+	+
Iqbal et al. (2021)	—	—	—	—	+	—	—	+	+
Jin et al. (2022)	—	?	?	?	+	—	?	—	+
Khan et al. (2022)	—	?	?	?	+	—	?	+	+
Kuang et al. (2019)	—	—	?	?	+	—	—	+	+
Lecumberri et al. (2013)	—	+	?	+	—	—	?	+	+
Li et al. (2023)	—	?	—	?	+	—	?	+	+
S. Li et al. (2021)	—	?	—	?	—	—	?	+	+
Li (2020)	—	?	?	?	+	—	—	+	+
Li et al. (2019)	—	?	?	?	+	—	?	+	+
Liu et al. (2020)	—	—	—	—	+	—	—	+	+
Macbeth et al. (2016)	—	+	?	+	—	—	?	+	+
Martin et al. (2016)	—	—	?	?	?	—	?	+	+
Michalski et al. (2023)	—	+	—	+	+	—	?	+	+
Munoz et al. (2023)	—	?	—	?	?	—	?	+	+
Muñoz Martín et al. (2018)	—	?	—	?	+	—	?	+	+
Niimi et al. (2024)	—	?	?	?	+	—	?	+	+
Oto et al. (2020)	—	?	?	?	—	—	?	+	+
Overvad et al. (2022)	—	—	—	—	+	—	—	+	+
Pabinger et al. (2018)	—	?	—	?	—	—	?	+	+
Pelzer et al. (2015)	—	+	—	+	—	—	?	+	+
Perry et al. (2010)	—	?	—	?	—	—	?	+	+
Qin et al. (2023b)	—	—	?	?	+	—	?	+	+
Qin et al. (2023a)	—	?	—	?	+	—	?	+	+
Ramos-Esquivel et al. (2023)	—	?	—	?	+	—	?	+	+
Rupa-Matysek et al. (2018)	—	—	—	—	+	—	—	+	+
Rupa-Matysek et al. (2017)	—	—	—	—	+	—	—	+	+
Sanfilippo et al. (2022)	—	?	—	?	+	—	?	+	+
Shibata et al. (2023)	—	?	—	?	+	—	?	+	+
Shoji et al. (2024)	—	?	?	?	+	—	?	+	+
Srikanthan et al. (2015)	—	—	?	?	+	—	—	+	+
Sun (2019)	—	?	?	?	—	—	?	+	+
Thipbunrung et al. (2023)	—	—	—	—	—	—	—	+	+
Trujillo-Santos et al. (2021)	—	?	?	?	—	—	?	+	+
Tsubata et al. (2023)	—	?	—	?	—	—	?	+	+
Van Es et al. (2017)	—	—	—	—	—	—	—	+	+
Vathiotis et al. (2018)	—	?	—	?	+	—	?	+	+
M.M. Wang et al. (2019)	—	?	?	?	+	—	?	+	+
X. Wang et al. (2019)	—	?	?	?	+	—	?	+	+
Yang et al. (2024)	—	?	?	?	+	—	?	—	+
Yang et al. (2021)	—	—	?	?	+	—	—	+	+
Yuen et al. (2022)	—	—	?	?	+	—	—	+	+
Zhang et al. (2022b)	—	—	?	?	+	—	?	+	+
Zhang et al. (2022a)	—	—	?	?	+	—	—	+	+
Zhang and Lin (2019)	—	?	—	?	+	—	?	+	+
Zhang (2018)	—	—	?	?	—	—	—	+	+
Zhang (2016)	—	+	—	+	—	—	—	+	+
Zhong (2021)	—	—	—	—	—	—	—	+	+

“+”: high risk of bias/high concern of applicability; “—”: low risk of bias/low concern of applicability; “?”: no information.

cancer types. For cancer types not mentioned above, the discriminative ability of the Khorana score remains unclear, and further studies are warranted to assess its discriminative ability.

It is noteworthy that the Khorana score was originally developed based on mixed cancer patients, including those with breast, colorectal, lung, lymphoma, gynecologic cancer, and other types of cancer

(Khorana et al., 2008), implying that it was intended to predict the risk of VTE across these cancers. However, our results showed that the Khorana score failed to accurately predict the risk of VTE in certain specific cancer types, particularly the aforementioned ones. One important reason for this may be that the risk factors for VTE are specific to different types of cancer, as confirmed by previous studies (Khorana et al., 2021; Sehrawat et al., 2021; Mulder et al., 2021; Timp et al., 2013), while the predictors included in the Khorana score are generic and do not reflect the characteristics of individual cancers. Therefore, on the one hand, we suggest developing specialized VTE risk prediction models for different types of cancer based on cancer-specific VTE risk factors to improve model prediction performance. The developer of the Khorana score has also made similar suggestions (Khorana et al., 2022). VTE risk prediction models for some specific cancer types have already been developed, such as for lung cancer (J. Li et al., 2021; Y. Li et al., 2021), breast cancer (Chen et al., 2022; Enajat et al., 2013), and multiple myeloma (Sanfilippo et al., 2022). On the other hand, we suggest that the Khorana score could be improved by incorporating certain cancer-specific VTE risk factors.

In addition to the aforementioned reason, other factors that might contribute to the poor discriminative ability of the Khorana score in external validation studies include heterogeneities in the study design, study population, clinical setting, and specific definition of predictors and outcomes between the original study and the external validation studies (Debray et al., 2017). Generally, the more similar the overall characteristics of the external validation studies are to those of the model development study, the more consistent the model's predictive performance will be in the external studies compared to the original development study. As previously stated, the Khorana score was developed in a prospective cohort study of ambulatory cancer patients undergoing chemotherapy, utilizing pre-chemotherapy measurements to predict the risk of VTE during the chemotherapy period. Our study found that the Khorana score is currently not limited to validation in its originally applicable setting or anti-cancer treatment stage, but has also been extensively validated in hospitalization settings and other anti-cancer treatment stages (i.e., surgery and unspecific treatment phases). It is reasonable to validate the Khorana score in different settings or populations to extend its application scope.

However, even in the included studies with overall characteristics similar to the original study developing the Khorana score (i.e., prospective cohort design, outpatient setting, chemotherapy phase) (Martín et al., 2016; Overvad et al., 2022; Thipbumrung et al., 2023; Van Es et al., 2017), the C-indices ranged from 0.52 to 0.60, indicating poor discriminative ability. Furthermore, regardless of whether subgroup analyses were conducted according to study design (observational study and RCT), cancer type (breast cancer, bladder cancer, urothelium cancer, glioma, gynecological cancer, pancreatic cancer, and mixed cancer), clinical setting (outpatient and hospitalization), or country (the United States, China, Spain, Netherlands, etc.), the discriminative ability of the Khorana score remained poor (pooled C-indices < 0.7). This finding indicates that in current clinical practice, the Khorana score may be unable to assist clinicians and nurses in accurately identifying cancer patients at high risk of VTE. However, more high-quality studies are needed to further evaluate its predictive performance, as the majority of the included studies were at high risk of bias. Additionally, most included studies were retrospective studies and used predictors assessed outside the pre-chemotherapy phase (the predictors of platelet, hemoglobin, and leukocyte were required to be tested before starting chemotherapy) and predicted the risk of VTE occurring outside the chemotherapy phase (the Khorana score was expected to predict the risk of VTE during the chemotherapy period). These factors also affected the validation results of the Khorana score.

Only four included studies reported results for another important indicator of model predictive performance, i.e., calibration. Calibration reflects the consistency between the model's predicted probabilities and the actual observed probabilities of the outcome, and it was most

commonly evaluated using a calibration plot (J.F. Wang et al., 2019). Although both discrimination and calibration are crucial for a risk prediction model, calibration is usually tested when discrimination is good. This may explain why most studies did not report calibration. This phenomenon is not limited to this systematic review but is also observed in other studies on systematic reviews of prediction models (Kunutsor et al., 2018; Lee et al., 2021). Studies tend not to report the results of model calibration, which may lead to reporting bias and hinder comprehensive validation of model prediction performance. Therefore, we suggest that future external validation studies should report calibration results regardless of whether the results are satisfactory, as this can help clinicians and nurses comprehensively evaluate the predictive performance of the model and make informed clinical decisions.

4.1. Implication for clinical practice

The Khorana score is strongly recommended by the American Society of Clinical Oncology (Key et al., 2023, 2020) and the Chinese Society of Clinical Oncology (Ma et al., 2019) for assessing the risk of VTE in patients with solid tumors receiving chemotherapy. Patients with a Khorana score of 2 or higher are categorized into high-risk groups and offered pharmacologic thromboprophylaxis. However, this systematic review found that the Khorana score might not accurately identify high-risk patients in clinical settings, suggesting it is time to reassess the role and effectiveness of this tool in predicting the risk of VTE in cancer patients. The Khorana score should be updated, or other potentially more accurate VTE risk prediction models should be explored and utilized to assist clinicians and nurses in preventing VTE more effectively.

In clinical practice, nurses use the Khorana score to evaluate patients' risk of VTE and deliver VTE prevention measures. Thus, it is also the nurses' responsibility to ensure that the chosen assessment tools are appropriate and accurate for the current clinical setting, which could improve nurses' critical thinking competence and ultimately benefit patients. The findings of this study alert nurses who use the Khorana score to assess cancer patients' VTE risk in clinical practice to pay attention to whether appropriate patients are receiving appropriate VTE prevention measures.

4.2. Strengths and limitations

One of the strengths of this systematic review is its comprehensiveness. We included 67 studies to assess the discrimination and calibration of the Khorana score, providing an updated perspective on its predictive performance. Additionally, we conducted subgroup analyses according to study design, clinical settings, cancer type, anti-cancer treatment phase, and country, which have not been comprehensively done in previous studies. These findings offer clinicians and nurses more valuable and accurate information when utilizing the Khorana score. Finally, weighting was applied to all analyses conducted in this systematic review based on the random effects model, enhancing the reliability of the study results.

This study also has several limitations. First, a significant limitation is the predominance of retrospective cohort or case-control studies included in the meta-analysis. Retrospective studies inherently carry biases such as selection bias, information bias, and recall bias, which can affect the reliability and validity of the findings, further impacting the quality of the evidence. As a result, the generalizability of our conclusions may be limited, and high-quality prospective studies are needed to robustly validate the predictive performance of the Khorana score. Second, due to the absence of 95 % CIs for the C-index in several included studies, the 95 % CIs for these studies were estimated based on recommended methods from previous studies (Hanley and McNeil, 1982; Steyerberg, 2019). Consequently, the results of the quantitative synthesis may have slight deviations. Finally, although our funnel plot and Egger test results did not show significant publication bias, we acknowledge that limiting our inclusion to studies published in English

and Chinese may lead to potential publication bias. This language restriction could result in the omission of relevant studies published in other languages, potentially affecting the comprehensiveness of our results. Therefore, future research should consider including studies published in multiple languages to further validate our findings and reduce potential publication bias. The abovementioned limitations highlight the need for caution when interpreting the results of this study.

5. Conclusions

The Khorana score has been currently widely evaluated in various cancer patients. However, its overall discrimination in cancer patients was limited, and the calibration was scarcely externally validated. Therefore, the Khorana score should be used cautiously, and further high-quality studies are needed to comprehensively validate its predictive performance.

Ethical approval

None.

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CRediT authorship contribution statement

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijnurstu.2024.104867>.

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